RuYi: Optimizing Burst Buffer Through Automated, Fine-Grained Process-to-BB Mapping

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***Abstract*—Current supercomputers use an SSD-based storage layer called Burst Buffer (BB) to provide I/O-intensive applications with accelerated storage access. However, efficiently utilizing this limited and expensive storage remains a critical issue, creating an urgent need for implementing Quality of Service (QoS) in BB. To address this, we propose RuYi, a QoS-aware method to provide applications with bandwidth guarantees in the BB file system. RuYi tackles two main issues. First, it quantitatively profiles available bandwidth resources in BB to ensure reliable QoS, a crucial aspect seldom studied in the literature. Second, RuYi offers fine-grained process-level QoS via an innovative process-to-BB mapping, maximizing resource utilization—something not achievable with conventional coarse-grained compute-to-BB mapping. We evaluated RuYi on a subsystem of the leading exascale supercomputer Sunway, consisting of 4,000 compute nodes and 200 BB nodes. The experimental results demonstrate that RuYi achieves an impressive end-to-end bandwidth control accuracy of 97%, while improving BB utilization by up to 116% compared to conventional coarse-grained compute-to-BB mapping.**

***Index Terms*—HPC, burst buffer, file system, QoS.**

I. Introduction

BURST Buffer (BB for short) [1] serves as an intermediate storage layer that sits between the compute nodes and the underlying parallel file system (PFS). Leveraging high-speed storage media such as SSDs, BB effectively absorbs bursty I/O operations and allows the applications to quickly return to compute stages. This in turn enhances the overall performance and reduces core usage hours [2]. BB has recently gained widespread adoption in high performance computing systems, scaling from petascale platforms such as Cori and Trinity to exascale systems such as Aurora and the new-generation Sunway.

Despite BB's appealing performance characteristics, its high cost makes it a scarce resource [3]. Therefore, sharing limited BB resources among multiple applications is inevitable. In this shared scenario, applications are susceptible to performance degradation due to interference from other applications. To achieve optimized BB allocation, existing research studies propose either interference-aware methods [4], [5], [6], [7], [8], [9] or QoS-aware methods [10], [11], [12], [13]. The former methods mainly detect inter-application interference and utilize efficient I/O sharing policies to maximize overall bandwidth, while the latter methods prioritize ensuring the bandwidth requirements of individual applications via resource reservation.

The dilemma between guaranteeing the bandwidth requirements of individual applications and maximizing overall BB utilization stems from two critical issues. First, existing methods lack the ability to quantitatively profile the available bandwidth on BB nodes. This capability is crucial and forms the foundation of almost all scheduling strategies. Without it, multiple applications can easily be allocated to BB nodes with insufficient resources, forcing them to share the limited resources. Second, current HPC systems typically adopt a compute-to-BB mapping where one compute node can be bound to only one BB node. The coarse-grained nature of the compute-to-BB mapping relied upon by most methods hinders fine-grained and flexible BB resource scheduling. For example, when a BB node has limited remaining resources, bandwidth demands at the granularity of compute nodes often cannot be fulfilled, resulting in wastage of resources on the BB node. These two critical issues motivate us to reexamine the design of the BB allocation method.

RuYi is a BB allocation mechanism designed for BB file systems to offer reliable and fine-grained I/O QoS. **To provide reliable QoS**, RuYi abstracts and maintains a dynamic bandwidth pool to quantitatively profile the available bandwidth resources in BB nodes. Specifically, since the consumption of SSD bandwidth resources fluctuates with the I/O characteristics of applications, RuYi requires three parameters, operation type (read or write), request size, and desired bandwidth, to conduct the profiling. **To provide process-level QoS**, RuYi adopts the process-to-BB mapping, allowing I/O processes of a compute node to be bound to different BB nodes. An I/O process refers to a process (such as a regular MPI process) that

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